

Biology

(A Level, Pearson Edexcel)

Edexcel A SNAB (Salters-Nuffield Advanced Biology) is largely taught in context through real-life biology. For example, we start with an account of cardio-vascular disease and then go on to look at the factors that make it more likely that any one of us will suffer from a stroke or heart attack. This allows us to introduce the biochemistry of fats and carbohydrates bit by bit, as you need to know them to understand about strokes and heart diseases, rather than all at once.

Some parts of the course are very practically based and you should be prepared to become involved in practical work to develop the necessary skills. Reading around the subject is also important as this will broaden your understanding.

Entry Requirements:

In Combined (Double) Science: Grade 6 in both exams.

In Separate (Triple) Sciences: Grade 6 in GCSE Biology, plus another 6 in another science.

Grade 5 in GCSE Mathematics.

Course Content and Assessment

Units	%	When?	Content
Paper 1	33.3	May/ June 2027	Topic 1: Lifestyle, health and risk This topic builds on students' knowledge and understanding of the functioning of the circulatory system and the importance of lifestyle choices to health. Topic 2: Genes and health This topic considers several biological principles related to cystic fibrosis. The topic also allows for discussion of the social and ethical issues surrounding genetic screening for genetic conditions. Topic 3: The voice of the genome This topic follows the development of multi-cellular organisms from single cells to complex individuals. Topic 4: Biodiversity and natural resources This topic focuses on biodiversity and the wealth of natural resources used by humans. Topic 5: On the wild side This topic covers ecosystems, climate change and our responsibilities as stewards of the environment. Topic 6: Infection, immunity and forensics This topic covers forensic pathology, bacteria and viruses, and the evolutionary battles that take place between invading pathogens and their hosts.
Paper 2	33.3	May/ June 2027	In addition to topics 1 - 4 above, the paper will also examine: Topic 7: Run for your life This topic is centred on the physiological adaptations which enable humans/animals to undertake strenuous exercise. Topic 8: Grey matter The nervous system, brain imaging and the regions of the brain are covered.
Paper 3	33.3	May/ June 2027	The paper will include synoptic questions drawn from two or more different topics from the specification. A pre-released scientific article will be available from Easter and this will be studied in class and at home. Approximately 30% of the paper will consist of comprehension questions based on the article.
Practical competency	Pass/Fail		Course endorsed practical skills are assessed continuously by the teacher(s) and only at the completion of the course is a pass or fail assessment submitted to the exam board.

All three papers will include questions that target the conceptual and theoretical understanding of experimental methods.

All three papers will include questions that target mathematics at higher level GCSE. This will equate to a minimum of 10% of the marks across all three papers. There is no longer any assessed coursework, but students will expect to demonstrate practical competency. There may be an opportunity to access University Level Laboratories in a field trip at the end of Year 12.

Career Links

The course provides a sound grounding in Biology for those wishing to pursue a variety of biologically based courses and careers. There is an emphasis on Human Biology that provides a good introduction for those interested in pursuing Medicine and related subjects.

For more information please contact Ms Howe